

Bayesian Modeling and MCMC Computation for Continuous Glucose Monitoring Time-Series Data

Revised implementation report aligned with the MCMC theory chapter

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Abstract

This report implements the Markov chain Monte Carlo principles developed in the companion theory report in a continuous glucose monitoring application. The empirical model combines a seasonal ARIMA conditional mean, Student- t innovations with an unknown degrees-of-freedom parameter, adaptive block Metropolis–Hastings during warmup, fixed-proposal sampling after warmup, modern multi-chain diagnostics, residual checks, and horizon-specific forecast evaluation. The analysis distinguishes computational reliability from statistical adequacy. Rank-normalized split $\hat{R}$, Bulk ESS, Tail ESS, Monte Carlo standard errors, trace plots, density plots, and rank plots are used to assess the retained posterior simulation. Residual diagnostics and an ARCH-LM test then examine features not explained by the constant-scale Student- t model. Predictive performance is evaluated over the final 48-hour holdout using a rolling multi-horizon design. Model-specific parameters are estimated from the training sample and then fixed, while the observed CGM history and residual state are updated after each new 15-minute observation. Forecasts are evaluated at horizons from 15 to 120 minutes. The report does not claim that Student- t innovations are new, that convergence has been proven, or that one model is universally best. Its contribution is a transparent and reproducible CGM-specific integration of model specification, posterior computation, diagnostics, and predictive assessment.

Contents

1	Scope, notation, and contribution	3
2	Data source, preparation and rolling forecast split	3
3	Preliminary motivation for Student-t innovations	4
4	SARIMA-t model	6
4.1	Conditional mean	6
4.2	Student- t innovations	6
4.3	Transformed parameters and prior	7
5	Adaptive block Metropolis–Hastings	7

6	Posterior constraints and root audit	9
7	Posterior summaries and MCMC reliability	9
7.1	Posterior summaries	9
7.2	Modern multi-chain diagnostics	10
7.3	Highest-density and equal-tail intervals	11
8	Residual diagnostics and remaining variance dynamics	14
9	Rolling multi-horizon predictive evaluation	16
9.1	Rolling forecast protocol	16
9.2	Predictive performance across horizons	16
9.3	Paired CRPS comparison and bootstrap uncertainty	20
9.4	Calibration diagnostics	20
10	Connection to the MCMC theory report	21
11	Limitations	21
12	Reproducibility manifest	22
13	Contributions and code provenance	23
14	Conclusion	23
	Declaration of generative AI and AI-assisted technologies	23
A	Remaining reproducibility checks	24
B	Superseded bootstrap specification	24
C	Superseded overlapping-target PIT calculation	24
	Bibliography	26

1 Scope, notation, and contribution

Two distinct sequences appear in this study. The observed CGM record is indexed by physical time $t = 1, \dots, N$. MCMC output is indexed by simulation iteration $m = 1, \dots, M$. The MCMC output is a Markov chain by construction. The physiological time series should not itself be described as a Markov chain unless an explicit conditional-independence model justifies that statement.

The index t denotes physical time and rolling forecast origin, m denotes MCMC iteration or retained draw, and h denotes forecast horizon in 15-minute steps. The vector $\boldsymbol{\eta}$ denotes unconstrained computational parameters, while $\boldsymbol{\theta}(\boldsymbol{\eta})$ denotes the corresponding natural-scale model parameters.

The methodological contribution is not the use of Student- t innovations alone. Heavy-tailed likelihoods and Bayesian Student- t models are established methods. The defensible contribution is the CGM-specific integration of:

1. 15-minute observations and a 96-period daily seasonal structure;
2. a SARIMA conditional-mean model;
3. Student- t innovations with unknown tail thickness;
4. adaptive block Metropolis–Hastings during warmup followed by fixed-proposal sampling;
5. modern multi-chain diagnostics and Monte Carlo precision assessment;
6. residual and conditional-variance diagnostics; and
7. horizon-specific predictive evaluation.

The analysis addresses five questions:

1. Are the residuals heavier-tailed than a Gaussian reference?
2. Are the retained MCMC estimates computationally reliable?
3. Does time-varying conditional variance remain after fitting the model?
4. How does forecast performance change with prediction horizon?
5. When does daily seasonal structure become useful?

2 Data source, preparation and rolling forecast split

The analysis uses a single participant record from the publicly available *ShanghaiT2DM* dataset, specifically `Shanghai_T2DM/2000_0_20201230.xlsx` [7, 8]. The record contains 1339 observations at 15-minute intervals, spanning 2020-12-30 13:54:00 to 2021-01-13 12:24:00 with no missing intervals; all successive timestamp differences equal 15 minutes.

Because the data are de-identified and publicly released by the original investigators, no additional ethics approval was sought for this secondary analysis. The original data collection was

approved by the Ethics Committees of Shanghai Fourth People’s Hospital and Shanghai East Hospital, both affiliated to Tongji University, in accordance with the Declaration of Helsinki, and informed consent was obtained from all participants [7]. Glucose was measured with a FreeStyle Libre H flash glucose monitoring device (Abbott Diabetes Care, Witney, UK), which stores a reading every 15 minutes.

Measurements are recorded on a 0.1 mmol/L grid, corresponding to a quantisation step of 1.8 mg/dL; the full series contains 103 distinct values. This resolution bounds the practical interpretation of differences between models (Section 11). The supplied CGM series contains 1339 observations recorded at 15-minute intervals. The first 1147 observations form the training sample. The final 192 observations form a fixed 48-hour holdout. Since one day contains 96 observations, the daily seasonal period is set to $s = 96$.

Let x_t denote CGM on the original mg/dL scale. The model is fitted to a centered series

$$y_t = x_t - \bar{x}_{\text{train}}, \quad (1)$$

where the training mean is computed using the training sample only. The same training mean is used to center the holdout. Forecasts are converted back to the original CGM scale exactly once after simulation. This convention prevents data leakage and avoids accidentally adding the training mean twice.

Table 1: Data split and reproducibility information.

Quantity	Value
Total observations	1339
Training observations	1147
Holdout observations	192
Sampling interval	15 minutes
Daily period	96 observations
Training-array SHA-256	29c93426280fc600ffa2be22dc3f127d917da6ee215343aa08895a699437f83d
Holdout-array SHA-256	78f4c5ec76e5407d1c3ceb2301a54041aae0ce0cedb38f11ac169569519e465e

Daily overlays and an average 24-hour profile are used as descriptive motivation for testing daily seasonality. Repeated timing patterns are visible, but the height and timing of peaks vary across days. This motivates a daily component without proving that one seasonal specification is uniquely correct. The absence of timestamp-matched meals, insulin doses, exercise, and sleep prevents causal attribution of individual peaks.

The holdout is evaluated sequentially. Forecast origins are restricted to the holdout. At each origin, the information set contains the complete training sample and all holdout observations revealed up to that origin. Model parameters remain fixed at their training-based estimates. After the next observation becomes available, it is added to the history, the residual state is updated, and the forecast origin advances by one step. For horizon h , the number of valid targets is $n_h = 192 - h + 1$.

3 Preliminary motivation for Student- t innovations

Before fitting the Bayesian Student- t model, Gaussian and Student- t reference distributions are compared using residuals from the same preliminary Gaussian SARIMA fit. Clearer tail depar-

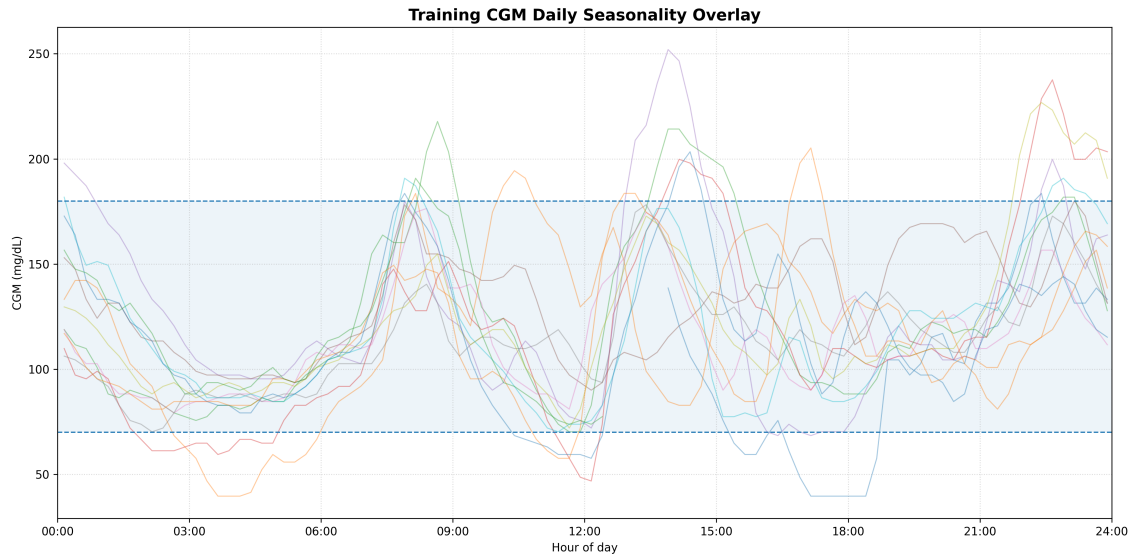


Figure 1: Daily overlays of the 1339 observed 15-minute CGM measurements. Repeated time-of-day structure motivates testing a daily seasonal period, while variation across days shows that the seasonal pattern is not deterministic.

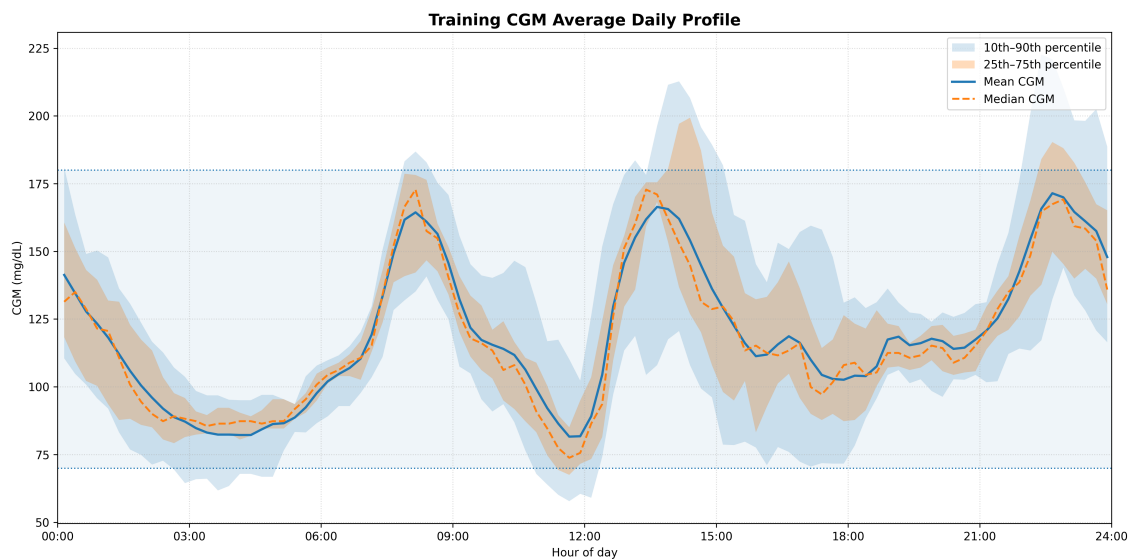


Figure 2: Average 24-hour CGM profile with a pointwise uncertainty band. The figure summarizes recurring daily timing while retaining visible uncertainty in the magnitude of the profile.

tures under the Gaussian reference provide descriptive motivation for a heavy-tailed likelihood. This preliminary comparison does not prove that the true innovation distribution is Student- t , and it does not establish predictive superiority.

A marginal distribution fit to fixed Gaussian-SARIMA residuals must not be interpreted as a full model comparison. The residuals remain fixed, and the Gaussian and Student- t SARIMA models are not separately re-estimated in that diagnostic. A valid matched comparison must hold the training set, holdout, conditional-mean structure, forecast origins, and scoring rules fixed.

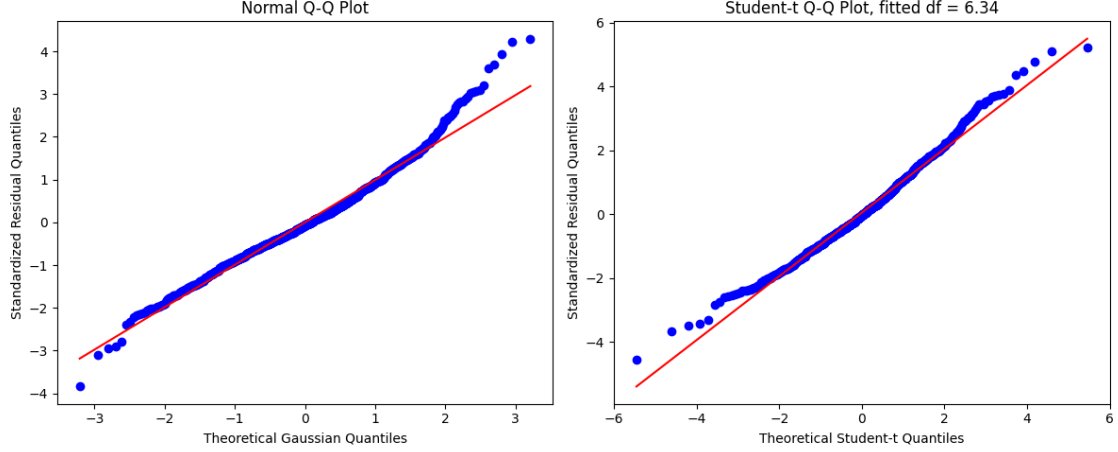


Figure 3: Preliminary Gaussian and Student- t Q-Q comparison based on the same fixed Gaussian-SARIMA residuals. This figure motivates a heavy-tailed innovation model; it is not a posterior diagnostic for the final Bayesian SARIMA- t fit.

4 SARIMA- t model

4.1 Conditional mean

The main conditional-mean specification is

$$(1 - \phi_1 B - \phi_2 B^2)(1 - \Phi_1 B^{96})y_t = (1 + \theta_1 B + \theta_2 B^2)\epsilon_t. \quad (2)$$

Expanding Equation 2 gives

$$\begin{aligned} \epsilon_t = & y_t - \phi_1 y_{t-1} - \phi_2 y_{t-2} - \Phi_1 y_{t-96} + \phi_1 \Phi_1 y_{t-97} + \phi_2 \Phi_1 y_{t-98} \\ & - \theta_1 \epsilon_{t-1} - \theta_2 \epsilon_{t-2}. \end{aligned} \quad (3)$$

The conditional residual array is initialized at zero before the largest lag. The likelihood begins at mathematical time $t = 99$, yielding $1147 - 98 = 1049$ training residuals. This is a conditional-likelihood treatment of the initial state.

4.2 Student- t innovations

Student- t likelihoods provide an established robust treatment of large residuals [2].

The innovations satisfy

$$\epsilon_t \mid \sigma, \nu \sim t_\nu(0, \sigma), \quad \sigma > 0, \quad \nu > 2. \quad (4)$$

Here σ is the Student- t scale parameter, not the innovation standard deviation. Under this parameterization,

$$\text{Var}(\epsilon_t \mid \sigma, \nu) = \sigma^2 \frac{\nu}{\nu - 2}. \quad (5)$$

The conditional log-likelihood is

$$\begin{aligned} \log L_{\text{Stud}} = \sum_{t=99}^{1147} & \left[\log \Gamma\left(\frac{\nu+1}{2}\right) - \log \Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2} \log(\pi \nu \sigma^2) \right. \\ & \left. - \frac{\nu+1}{2} \log\left(1 + \frac{\epsilon_t^2}{\nu \sigma^2}\right) \right]. \end{aligned} \quad (6)$$

4.3 Transformed parameters and prior

Define

$$\boldsymbol{\eta} = (\phi_1, \phi_2, \theta_1, \theta_2, \Phi_1, \eta_\sigma, \eta_\nu), \quad (7)$$

with

$$\sigma = \exp(\eta_\sigma), \quad \nu = 2 + \exp(\eta_\nu). \quad (8)$$

The prior places independent normal distributions on the transformed parameters:

$$\phi_1, \phi_2, \theta_1, \theta_2 \sim \mathcal{N}(0, 1.5^2), \quad (9)$$

$$\Phi_1 \sim \mathcal{N}(0, 1.0^2), \quad (10)$$

$$\eta_\sigma = \log \sigma \sim \mathcal{N}(\log 10, 1.0^2), \quad (11)$$

$$\eta_\nu = \log(\nu - 2) \sim \mathcal{N}(\log 6, 1.0^2). \quad (12)$$

The implied prior medians are $\sigma = 10$ and $\nu = 8$. The implied central 95% prior interval for ν is approximately $[2.85, 44.6]$, which is substantially wider than the posterior 95% HDI $[4.158, 9.546]$, indicating that the data rather than the prior drive the posterior for tail thickness. A hard constraint $|\Phi_1| < 0.999$ is imposed inside the target; draws violating it receive $-\infty$ log-density.

The prior is defined directly on $\boldsymbol{\eta}$. Therefore, the implemented posterior target is

$$\log \pi_{\boldsymbol{\eta}}(\boldsymbol{\eta} \mid \mathbf{y}) = \log L_{\text{Stud}}\{\mathbf{y} \mid \boldsymbol{\theta}(\boldsymbol{\eta})\} + \log \pi_{\boldsymbol{\eta}}(\boldsymbol{\eta}), \quad (13)$$

and no additional Jacobian term is required. A Jacobian would be required only if a prior were first defined on (σ, ν) and then expressed in transformed coordinates.

5 Adaptive block Metropolis–Hastings

All seven transformed parameters are proposed jointly:

$$\boldsymbol{\eta}^* \sim \mathcal{N}\left(\boldsymbol{\eta}^{(m-1)}, \boldsymbol{\Sigma}_m\right). \quad (14)$$

The empirical-covariance adaptation follows the adaptive Metropolis framework [3], with $2.38^2/d$ used as a random-walk scaling reference [4].

During warmup, the proposal covariance is updated using the empirical covariance of the chain:

$$\Sigma_m = \frac{2.38^2}{d} \widehat{\text{Cov}}_m(\boldsymbol{\eta}) + 10^{-6} I_d, \quad d = 7. \quad (15)$$

Adaptation begins at iteration 500 and occurs every 50 iterations. The final update occurs at iteration 3950. From iteration 4000 onward, the proposal covariance remains fixed. Posterior summaries use only the fixed-proposal phase. During retained sampling, $\Sigma_m = \Sigma_{\text{final}}$ for every post-warmup iteration.

Table 2: Official MCMC configuration.

Setting	Value
Number of chains	4
Iterations per chain	12000
Warmup iterations per chain	4000
Last adaptation iteration	3950
Post-warmup iterations per chain	8000
Thinning factor	4
Retained draws per chain	2000
Total retained draws	8000

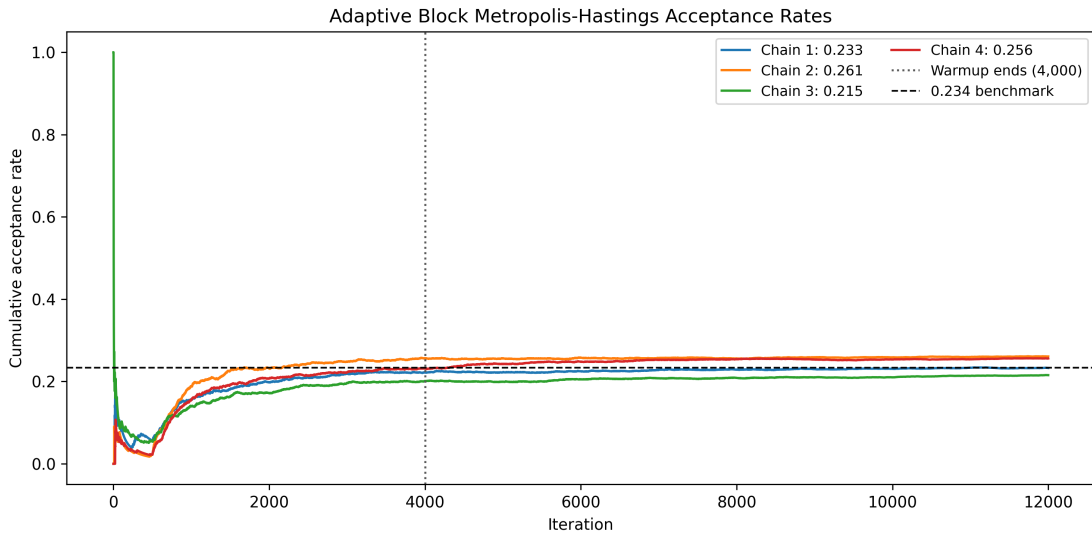


Figure 4: Cumulative acceptance rates for the four adaptive block Metropolis–Hastings chains. The vertical boundary separates warmup from the fixed-proposal retained phase. The 0.234 line is a tuning reference, not a convergence threshold.

Thinning is used only for storage and downstream computational convenience. It is not treated as a substitute for autocorrelation diagnostics. Sampling efficiency is assessed using Bulk ESS, Tail ESS, and MCSE.

Stopping adaptation gives the retained phase a homogeneous proposal mechanism, but stopping adaptation does not prove convergence. The fixed random-walk proposal has nonzero support in every direction, and the MH acceptance rule preserves the implemented posterior

target. Adequate exploration remains an empirical question assessed using multiple chains and diagnostics.

6 Posterior constraints and root audit

The current implementation explicitly constrains the seasonal coefficient through $|\Phi_1| < 0.999$. The custom target does not enforce the joint nonseasonal AR stationarity region or the MA invertibility region during sampling. A post hoc audit nevertheless found that all retained draws satisfied the AR(2) stationarity and MA(2) invertibility conditions. This result supports the retained sample, but it is not equivalent to using a constraint-preserving parameterization.

For each posterior draw, define the AR polynomial

$$a(z) = 1 - \phi_1 z - \phi_2 z^2 \quad (16)$$

and the MA polynomial

$$b(z) = 1 + \theta_1 z + \theta_2 z^2. \quad (17)$$

The post hoc implementation audit reports the proportion of retained draws for which all roots of both polynomials lie outside the unit circle.

Table 3: Post hoc stationarity and invertibility audit of the retained Student- t SARIMA draws.

Quantity	Result
Posterior draws satisfying AR(2) stationarity	100.00%
Posterior draws satisfying MA(2) invertibility	100.00%
Median minimum AR root modulus	1.2812
Median minimum MA root modulus	1.6967

The minimum observed AR and MA root moduli were 1.1156 and 1.3432, respectively; the corresponding 2.5th percentiles were 1.1679 and 1.4554. The AR roots are outside the unit circle for all retained draws, but they are not far from the stationarity boundary. In particular, the posterior means imply $\phi_1 + \phi_2 \approx 0.90$, and the median minimum AR-root modulus is only 1.2812. This indicates a highly persistent autoregressive component, so long-horizon forecasts are relatively sensitive to uncertainty in the AR coefficients. Such persistence is consistent with the gradual decline in predictive R^2 observed in Figure 10, Panel F, as the forecast horizon increases. Although every retained draw passed the post hoc audit, a future implementation should enforce these regions directly through a root-based rejection rule or a constraint-preserving transformation.

7 Posterior summaries and MCMC reliability

7.1 Posterior summaries

The posterior supports short-run autoregressive dependence and positive dependence on the same time of the previous day. The 95% HDI for θ_2 includes zero, so the direction and magnitude of that individual effect remain uncertain. The posterior for ν is concentrated at moderate values

Table 4: Posterior summaries from the fixed-after-warmup run. Means and 95% HDIs are computed after transforming each retained draw to the natural parameter scale.

Parameter	Posterior mean	95% HDI
ϕ_1	1.306	[1.153, 1.465]
ϕ_2	-0.409	[-0.557, -0.264]
θ_1	0.779	[0.609, 0.951]
θ_2	0.111	[-0.039, 0.260]
Φ_1	0.186	[0.128, 0.242]
σ	5.058	[4.675, 5.422]
ν	6.777	[4.158, 9.546]

rather than near the Gaussian limiting case. This supports heavier-tailed innovations within the fitted model, but it does not establish predictive superiority over Gaussian SARIMA.

7.2 Modern multi-chain diagnostics

Following modern MCMC recommendations [5], the retained draws are evaluated using rank-normalized split $\hat{R}$, Bulk ESS, Tail ESS, MCSE of posterior means, quantile-specific MCSE, trace plots, chain-specific density plots, and rank plots. These measures answer different questions. $\hat{R}$ checks compatibility across chains, ESS assesses sampling efficiency, and MCSE assesses numerical precision.

Table 5: Summary of modern MCMC diagnostics.

Diagnostic	Observed range or value
Rank-normalized split $\hat{R}$	1.001–1.003
Bulk ESS	1293–1472
Tail ESS	1714–2440
MCSE of posterior mean for ν	0.041
Largest MCSE(mean)/posterior SD	2.83% (for ν)

The diagnostics provide no clear computational warning for the monitored parameters. This finding supports numerical stability among the chains that were run, but it does not prove that every relevant region of the posterior distribution was explored. The interpretation of $\hat{R}$ also depends on the initialization of the chains. The four chains were initialized from a common preliminary reference vector $\boldsymbol{\eta}_0$ on the unconstrained parameter scale. This reference vector was constructed from the training-sample Gaussian SARIMAX estimates:

$$\boldsymbol{\eta}_0 = (1.208645, -0.323417, 0.876772, 0.190607, 0.209988, 1.850369, 1.791759). \quad (18)$$

The sixth and seventh elements are

$$\eta_\sigma = \log(\sqrt{40.477156}) = 1.850369$$

and

$$\eta_\nu = \log(8 - 2) = \log 6 = 1.791759.$$

They correspond to the natural-scale initial values $\sigma_0 = \sqrt{40.477156} \approx 6.362$ and $\nu_0 = 8$. For chain $c = 0, 1, 2, 3$, the initial value was generated as

$$\boldsymbol{\eta}_c^{(0)} = \boldsymbol{\eta}_0 + \boldsymbol{\delta}_c, \quad (19)$$

where

$$\boldsymbol{\delta}_c \sim \mathcal{N}[\mathbf{0}, \text{diag}(0.03^2, 0.03^2, 0.03^2, 0.03^2, 0.02^2, 0.03^2, 0.08^2)]. \quad (20)$$

Independent chain-specific random-number generators used the base seed 20260720 plus the chain index c . The resulting chain seeds were therefore 20260720, 20260721, 20260722, and 20260723.

Relative to the posterior standard deviations on the corresponding sampling scales, the initialization perturbation standard deviations are approximately 37%, 40%, 34%, 39%, and 67% for ϕ_1 , ϕ_2 , θ_1 , θ_2 , and Φ_1 , respectively. The chains therefore started at distinct but still substantially overlapping locations within the posterior bulk, rather than from strongly overdispersed starting points. Consequently, the rank-normalized split $\hat{R}$ values primarily provide evidence of agreement among the chains that were run. They may have limited ability to reveal a separated posterior mode that was not reached by any chain.

ESS and MCSE are estimand-specific. The reported values apply to the monitored parameters and posterior summaries rather than to every possible posterior functional. The largest ratio of MCSE(mean) to posterior standard deviation was 2.83% for ν . The corresponding mean-specific effective sample size implied by the unrounded ratio was approximately 1252, compared with a rank-normalized Bulk ESS of approximately 1293 for the same parameter. Although these quantities are not defined identically, their similar magnitudes provide a useful numerical consistency check.

7.3 Highest-density and equal-tail intervals

The main posterior table reports 95% HDIs. Quantile-specific MCSE applies to equal-tail posterior quantiles and must not be attached to an HDI endpoint. The two interval types can differ for an asymmetric posterior.

Table 6: Equal-tail quantiles and quantile-specific Monte Carlo errors.

Parameter	2.5% quantile	MCSE at 2.5%	97.5% quantile	MCSE at 97.5%
ϕ_1	1.1432	0.0053	1.4566	0.0038
ϕ_2	-0.5510	0.0039	-0.2556	0.0057
θ_1	0.6054	0.0054	0.9488	0.0036
θ_2	-0.0429	0.0050	0.2588	0.0023
Φ_1	0.1292	0.0010	0.2436	0.0013
σ	4.6929	0.0114	5.4464	0.0192
ν	4.6311	0.0399	10.3172	0.1441

The upper equal-tail quantile of ν is less precisely estimated because it lies in the longer right tail. Its quantile MCSE does not refer to the upper endpoint of the HDI in Table 4.

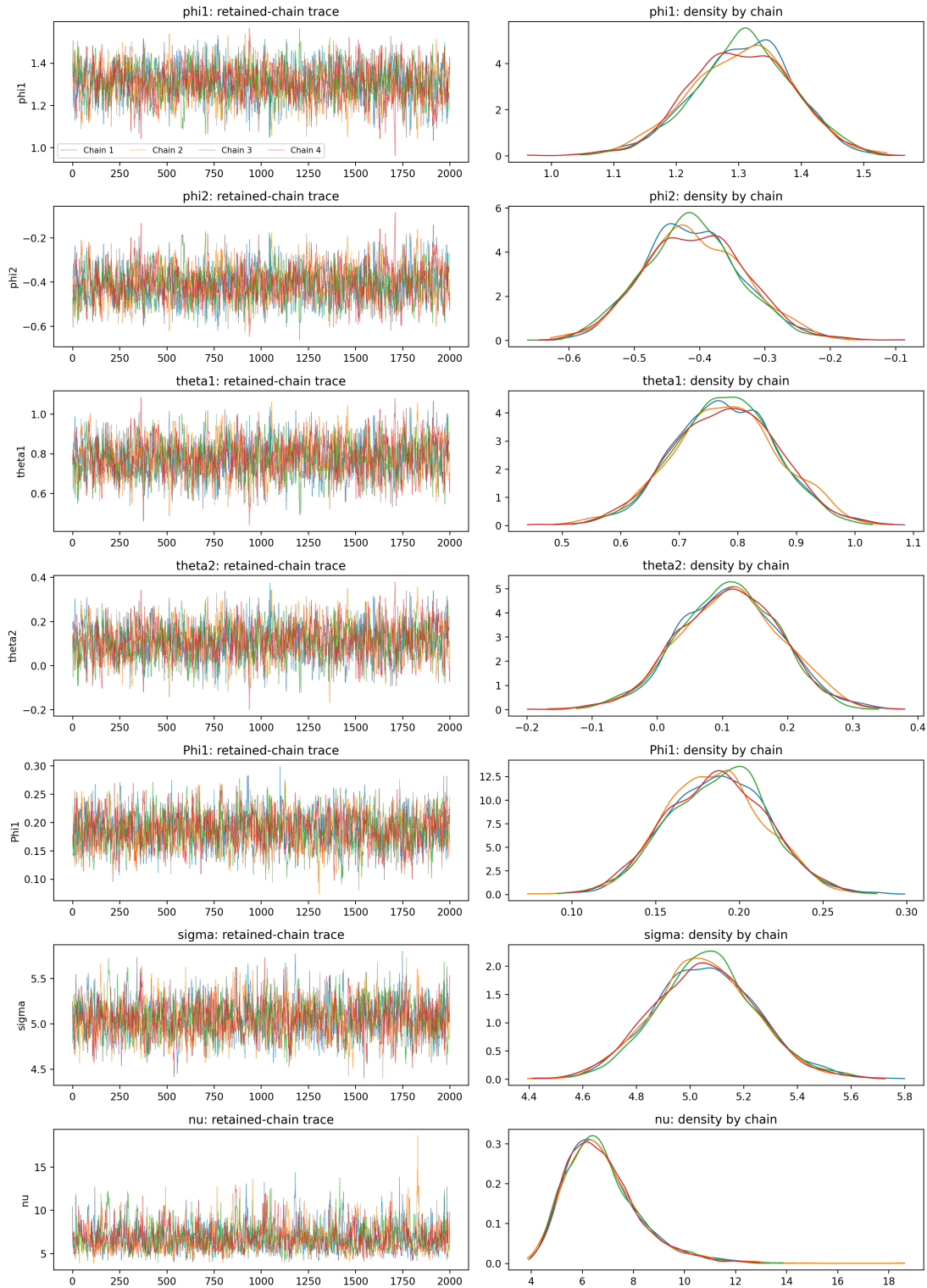


Figure 5: Retained-chain trace plots and chain-specific posterior densities for the fitted parameters. These graphical diagnostics complement the rank-normalized split $\hat{R}$, Bulk ESS, Tail ESS, and MCSE summaries. They are not treated as stand-alone evidence of convergence or complete posterior exploration.

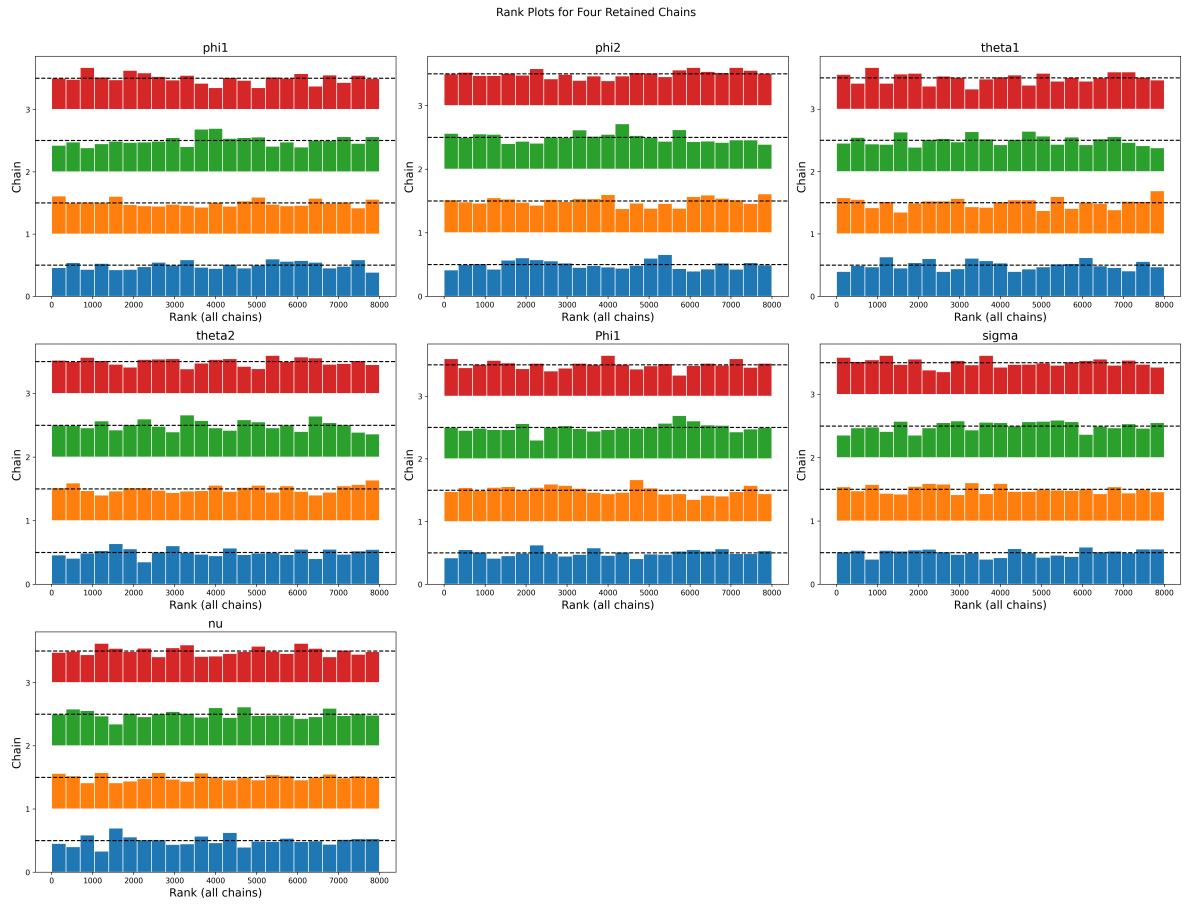


Figure 6: Rank plots for the four retained chains. Broadly similar rank contributions across chains are consistent with the numerical multi-chain diagnostics. Pronounced chain-specific concentration in particular rank ranges would indicate disagreement or incomplete mixing.

8 Residual diagnostics and remaining variance dynamics

Several large positive and negative residuals remain after fitting the model. Compared with a Gaussian likelihood, the Student- t likelihood reduces the relative influence of large residuals during estimation. It does not make every abrupt glucose excursion predictable.

The residual diagnostics include the residual sequence, a residual-density comparison, a Student- t Q-Q plot, and the residual ACF through lag 120. The Q-Q plot is broadly consistent with a Student- t reference, although deviations remain at the most extreme observations.

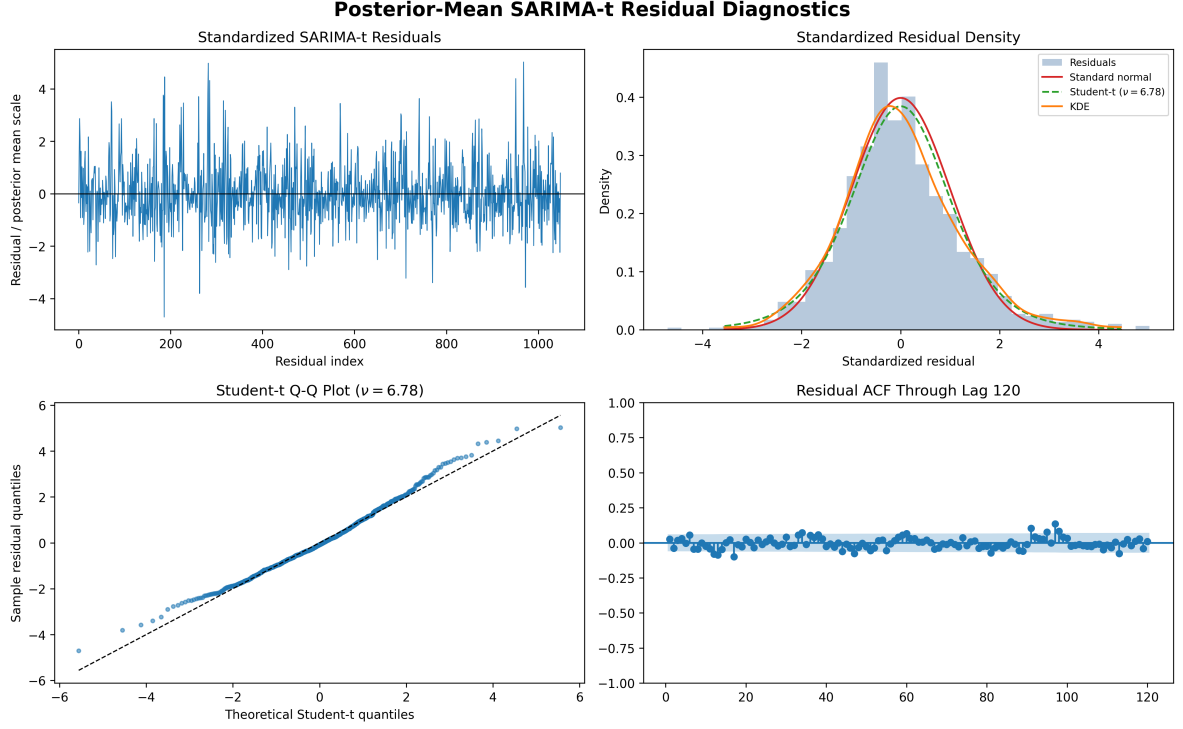


Figure 7: Posterior-mean SARIMA- t residual diagnostics, including the residual sequence, density comparison, Student- t Q-Q plot, and residual ACF. The overall distribution is broadly compatible with a heavy-tailed reference, although extreme observations and remaining scale dynamics are still visible.

Following Engle [1], an ARCH-LM auxiliary regression with 12 lags is applied to the posterior-mean residuals:

$$\hat{\epsilon}_t^2 = \alpha_0 + \sum_{j=1}^{12} \alpha_j \hat{\epsilon}_{t-j}^2 + u_t. \quad (21)$$

Table 7: ARCH-LM diagnostic for posterior-mean SARIMA- t residuals.

Test	Lags	Statistic	p -value
ARCH-LM	12	71.087	2.004×10^{-10}

The result provides strong evidence against constant conditional variance for the specified auxiliary regression. It motivates a time-varying scale model, but it does not uniquely identify GARCH. Stochastic volatility, regime-dependent scale, time-of-day effects, covariate-dependent variance, and omitted clinical events are also plausible extensions.

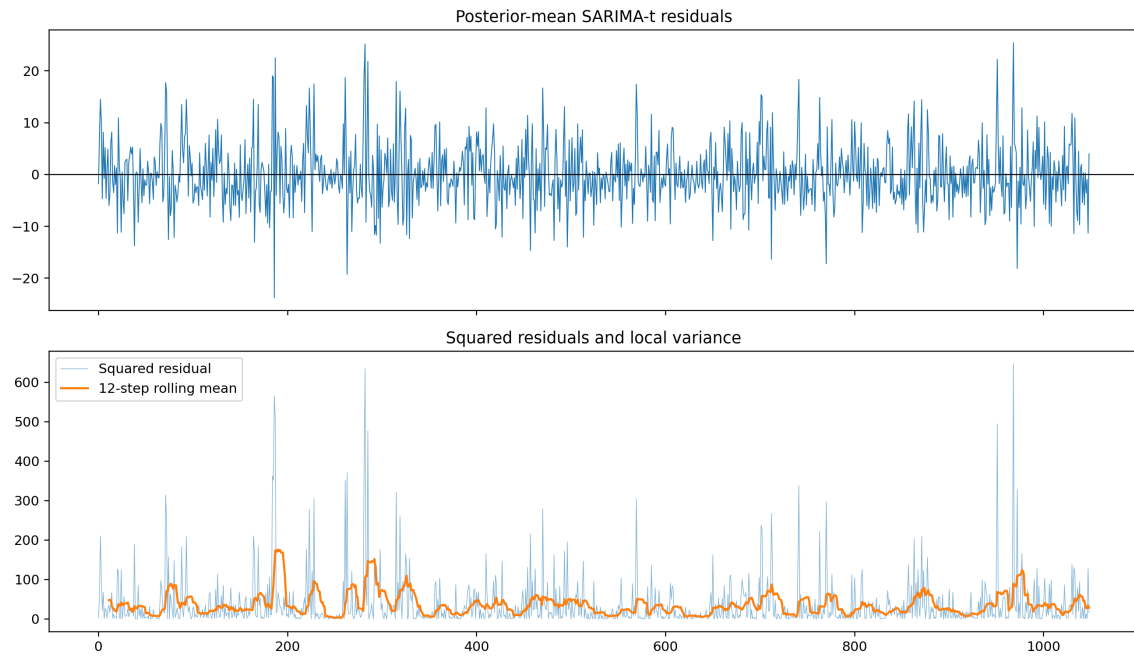


Figure 8: Posterior-mean residuals, squared residuals, and the rolling mean of squared residuals. Clustering in squared residuals provides a visual complement to the ARCH-LM test.

The plug-in test does not propagate posterior parameter uncertainty. A fuller Bayesian diagnostic could compute residual-scale summaries across posterior draws or compare replicated squared-residual patterns in posterior predictive checks.

9 Rolling multi-horizon predictive evaluation

9.1 Rolling forecast protocol

The final 192 observations define a 48-hour rolling evaluation period. Each model is estimated using only the first 1147 observations, and its parameter estimates remain fixed throughout the holdout. Let $\mathcal{I}_t$ denote information available at rolling origin t . For model j and horizon

$$h \in \{1, 2, 4, 6, 8\},$$

corresponding to 15, 30, 60, 90, and 120 minutes, the plug-in predictive distribution is

$$p_j(y_{t+h} \mid \mathcal{I}_t, \hat{\boldsymbol{\theta}}_j). \quad (22)$$

After the next CGM observation becomes available, the observed history and residual state are updated and the origin advances by one 15-minute step. No observation after the current origin is used to generate a forecast. For the Bayesian Gaussian and Student- t SARIMA models, $\hat{\boldsymbol{\theta}}_j$ is the posterior mean vector. For frequentist comparison models, it is the training-sample maximum-likelihood estimate. Innovation uncertainty is represented, but full posterior parameter uncertainty is not propagated.

The detailed 15-minute diagnostic and the multi-horizon comparison serve different purposes. Figure 9 emphasizes one-step predictive behavior, whereas Table 8 and Figure 10 summarize performance across forecast horizons. An earlier version of this work included a third specification combining a $K = 10$ Fourier daily mean with an ARIMA(2,0,3) Student- t error process. That specification is excluded here. Its 28-dimensional parameter vector was sampled with the same random-walk Metropolis–Hastings proposal used for the seven-dimensional models, and its convergence diagnostics were not satisfactory. It is not reported as a competitor because inadequate posterior computation cannot be distinguished from inadequate model fit. Re-estimating that specification with a gradient-based sampler is left to future work.

9.2 Predictive performance across horizons

Across all specifications, RMSE and CRPS increase with horizon, while explained holdout variation declines. The four models show little separation by CRPS at 15 and 30 minutes. This does not mean that model structure is unimportant at short horizons: a random-walk forecast using only the most recent observation attains an RMSE of 10.443 mg/dL at 15 minutes, whereas all four fitted models attain 5.86–5.98 mg/dL, an improvement of roughly 43%. The absence of separation therefore indicates that the four specifications recover essentially the same short-run dynamics, not that those dynamics are uninformative. The measurement resolution provides a physical bound on how finely models can be separated. CGM values are recorded on a 0.1 mmol/L grid, a quantisation step of 1.8 mg/dL, and the full series contains only 103 distinct values. At the 15-minute horizon the four models differ in RMSE by 0.023–0.127 mg/dL, that is by 1.3% to 7.1% of a single quantisation step. Differences of this size are not physically interpretable, independent of any statistical test.

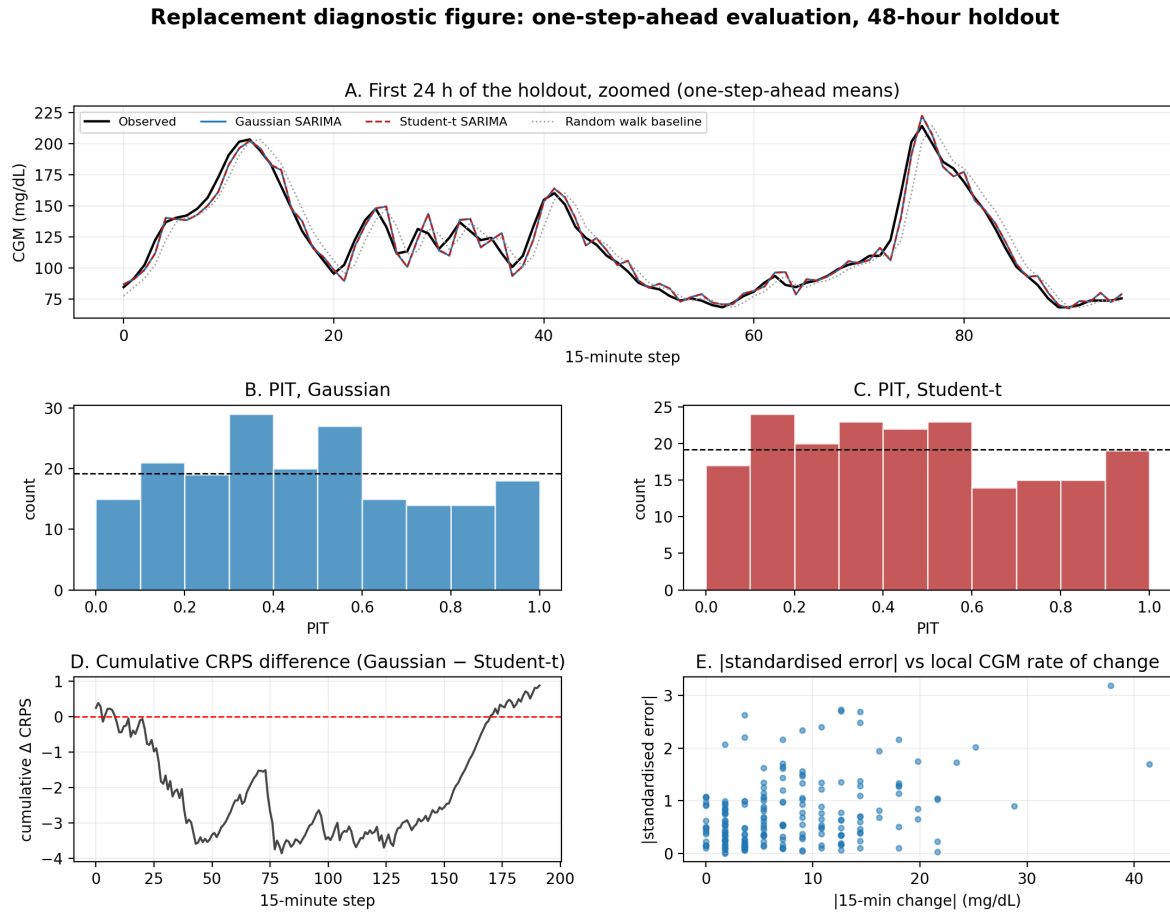


Figure 9: Fifteen-minute-ahead rolling predictive diagnostics over the final 48-hour holdout. All six panels use a one-step (15-minute) horizon. Parameters are estimated from the training sample and held fixed, while the observed history and residual state are updated at each origin. Panel A shows the first 24 hours of the holdout with the one-step conditional means of the Gaussian and Student- t models and a random-walk reference. Panels B and C show PIT histograms. Panel D shows the cumulative CRPS difference between the two Bayesian models. Panel E plots the absolute standardised error against the local 15-minute rate of change. Multi-horizon results appear in Figure 10.

Table 8: Rolling plug-in predictive performance by horizon. Model-specific parameters are fixed at their training-sample estimates, while the observed history and residual state are updated sequentially. Calibration is reported separately in Section 9.4, using non-overlapping targets.

Model	Horizon	RMSE	CRPS	95% coverage	95% width	R^2
Gaussian SARIMA	15 min	5.935	3.279	0.932	23.603	0.9717
Gaussian SARIMA	30 min	14.085	7.738	0.937	55.167	0.8405
Gaussian SARIMA	60 min	24.560	13.569	0.947	97.517	0.5150
Gaussian SARIMA	90 min	30.204	16.908	0.936	115.709	0.2665
Gaussian SARIMA	120 min	32.722	18.380	0.924	122.531	0.1391
Student- t SARIMA	15 min	5.958	3.274	0.932	24.001	0.9715
Student- t SARIMA	30 min	14.096	7.711	0.937	55.201	0.8402
Student- t SARIMA	60 min	24.562	13.541	0.942	96.965	0.5149
Student- t SARIMA	90 min	30.243	16.926	0.936	115.249	0.2646
Student- t SARIMA	120 min	32.829	18.432	0.930	122.451	0.1334
SARIMA MLE	15 min	5.983	3.316	0.943	24.858	0.9712
SARIMA MLE	30 min	14.168	7.807	0.948	57.418	0.8386
SARIMA MLE	60 min	24.723	13.659	0.947	99.315	0.5085
SARIMA MLE	90 min	30.344	16.945	0.941	116.923	0.2597
SARIMA MLE	120 min	32.857	18.405	0.930	123.948	0.1319
ARMA(2,2), no seasonal	15 min	5.856	3.252	0.953	25.334	0.9724
ARMA(2,2), no seasonal	30 min	14.140	7.767	0.958	59.449	0.8392
ARMA(2,2), no seasonal	60 min	25.450	14.091	0.952	105.795	0.4792
ARMA(2,2), no seasonal	90 min	32.029	17.998	0.947	126.387	0.1752
ARMA(2,2), no seasonal	120 min	35.348	19.962	0.930	134.872	-0.0047
Random walk (baseline)	15 min	10.443	—	—	—	0.9123
Random walk (baseline)	30 min	19.352	—	—	—	0.6989
Random walk (baseline)	60 min	32.793	—	—	—	0.1353
Random walk (baseline)	90 min	42.846	—	—	—	-0.4761
Random walk (baseline)	120 min	50.613	—	—	—	-1.0598
Seasonal naive (baseline)	15 min	36.164	—	—	—	-0.0516
Seasonal naive (baseline)	30 min	35.669	—	—	—	-0.0230
Seasonal naive (baseline)	60 min	34.734	—	—	—	0.0299
Seasonal naive (baseline)	90 min	34.619	—	—	—	0.0363
Seasonal naive (baseline)	120 min	34.761	—	—	—	0.0284

Table 9: Forecast target counts when forecast origins are restricted to the holdout period. For horizon h measured in 15-minute steps, $n_h = 192 - h + 1$. An alternative design that allows origins to begin before the holdout was not used, because it would evaluate forecasts partly on training observations. Non-overlapping PIT diagnostics retain every h th target.

Horizon	Rolling targets $n_h = 192 - h + 1$	Non-overlapping PIT targets
15 minutes	192	192
30 minutes	191	96
60 minutes	189	48
90 minutes	187	32
120 minutes	185	24

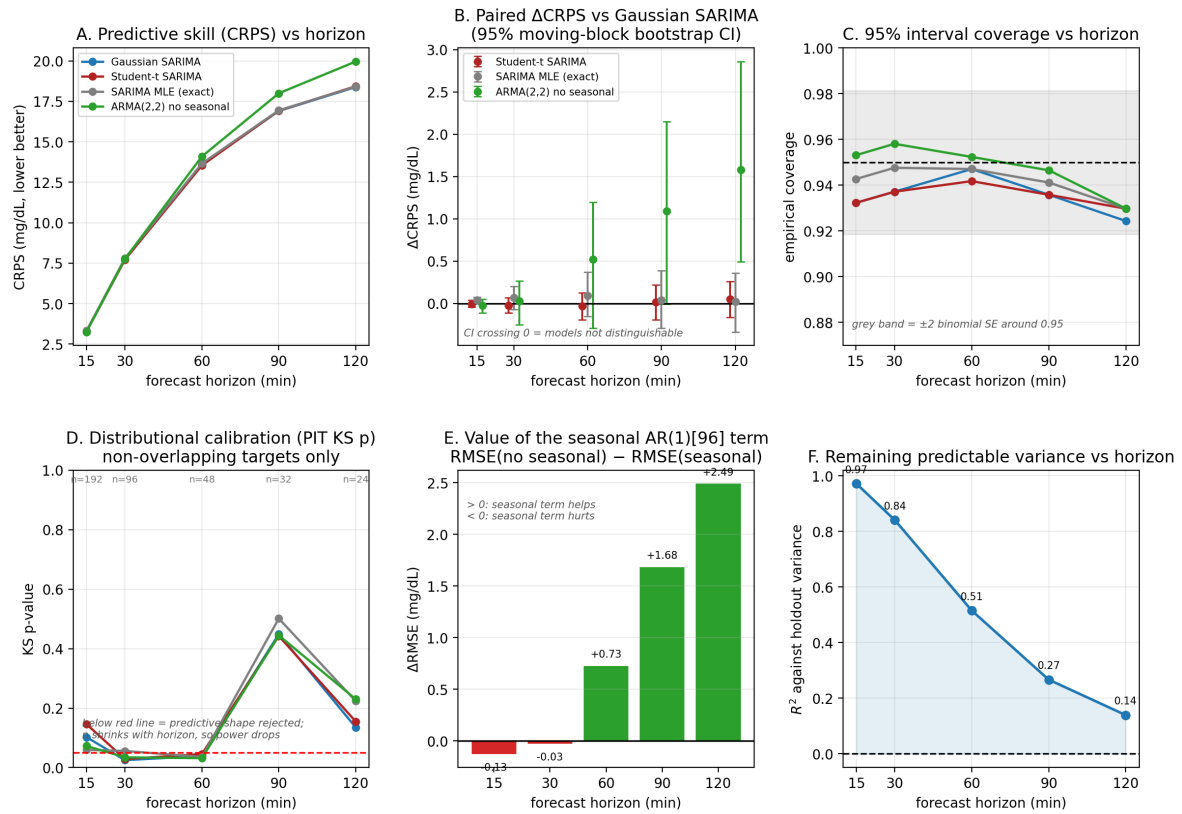
One-step to two-hour predictive comparison, 192-point holdout, fixed parameters, common random numbers

Figure 10: Rolling-origin predictive comparison over the 192-point, 48-hour holdout, with model-specific parameters fixed at their training-sample estimates and common random numbers used across models. Panel A shows that CRPS increases from approximately 3.3 mg/dL at 15 minutes to 18–20 mg/dL at 120 minutes, with little separation among the four models at 15 and 30 minutes. Panel B reports paired CRPS differences relative to Gaussian SARIMA with 95% moving-block-bootstrap intervals. Fourteen of the fifteen intervals include zero; the exception is the nonseasonal ARMA model at 120 minutes, with a CRPS difference of 1.58 mg/dL (95% CI: 0.49 to 2.86). Panel C shows that empirical 95% coverage generally declines at longer horizons. Panel D assesses PIT uniformity using non-overlapping forecast targets. Panel E shows that the daily seasonal AR term provides little or slightly negative benefit at 15–30 minutes but becomes beneficial from 60 minutes onward. Panel F shows that explained holdout variation declines from approximately $R^2 = 0.97$ at 15 minutes to $R^2 = 0.14$ at 120 minutes.

9.3 Paired CRPS comparison and bootstrap uncertainty

For model j , horizon h , and rolling origin t , define

$$d_{j,h,t} = \text{CRPS}_{j,h,t} - \text{CRPS}_{G,h,t}, \quad (23)$$

where G denotes Gaussian SARIMA. The mean paired difference is

$$\bar{d}_{j,h} = \frac{1}{n_h} \sum_t d_{j,h,t}. \quad (24)$$

Positive values indicate higher, and therefore worse, CRPS than Gaussian SARIMA.

The horizon-focused analysis in Figure 10 uses a moving-block percentile bootstrap applied to consecutive paired rolling-origin loss differences. Its verified settings are a block length of 12 forecast origins, corresponding to three hours, and 4000 bootstrap replications. Common random numbers are used across models through a common uniform array indexed by forecast origin, horizon, recursive simulation step, and simulation index. Each model transforms the shared uniforms through its own innovation inverse CDF. This alignment reduces Monte Carlo noise in paired differences without imposing identical predictive distributions.

Fourteen of the fifteen model-by-horizon intervals include zero. The exception is the nonseasonal ARMA model at 120 minutes, whose mean CRPS is 1.58 mg/dL higher than Gaussian SARIMA (95% CI: 0.49 to 2.86 mg/dL). No multiplicity adjustment is applied, so these comparisons are interpreted as exploratory.

Table 10: Bootstrap specification used for Figure 10.

Setting	Value
Bootstrap method	Moving-block percentile bootstrap
Resampled quantity	Consecutive paired rolling-origin CRPS differences
Block length	12 forecast origins (3 hours)
Bootstrap replications	4000
Confidence level	95%
Common random numbers	Yes; shared uniform array indexed by origin, horizon, simulation step, and draw
Multiplicity adjustment	None; comparisons are exploratory

9.4 Calibration diagnostics

Empirical interval coverage is calculated using all available rolling forecast targets. PIT uniformity is assessed separately using non-overlapping targets because the standard Kolmogorov–Smirnov reference distribution does not account for the strong dependence created by overlapping multi-step forecasts.

For horizon h , every h th forecast target is retained for the PIT diagnostic. The resulting sample sizes are 192, 96, 48, 32, and 24 for horizons of 15, 30, 60, 90, and 120 minutes, respectively. For Gaussian SARIMA, the corrected non-overlapping KS p -values are approximately 0.102, 0.025, 0.039, 0.449, and 0.135 across the five horizons. The 30- and 60-minute results indicate mild calibration departures. The larger p -values at 90 and 120 minutes should not be interpreted as strong evidence of calibration because only 32 and 24 non-overlapping targets remain.

Across models, empirical coverage generally declines at longer horizons. Together with PIT summaries below 0.5 at intermediate horizons, this suggests predictive distributions that are slightly high-centered or too narrow. The pattern is consistent with, but not uniquely explained by, the residual evidence against constant conditional variance.

10 Connection to the MCMC theory report

Table 11: Mapping from the theory report to the CGM implementation.

Theoretical principle	Current implementation	Remaining requirement
Unnormalized log posterior	Student- t conditional likelihood plus prior on η	Preserve one version-controlled target
Multiple chains	Four chains	Document dispersed initial values
Warmup is not convergence	Warmup excluded from posterior summaries	Retain diagnostic wording that avoids claiming proof
Adaptive proposal tuning	Empirical covariance during warmup	Document fixed proposal in retained phase
Rank-normalized split $\hat{R}$	Reported for model parameters	Add relevant derived quantities
Bulk and Tail ESS	Reported for model parameters	Add roots and selected forecast functionals
MCSE	Mean and quantile MCSE reported	Add simulation uncertainty for forecast metrics
Graphical checks	Trace, density, rank, residual plots	Include representative plots in the report appendix
Model adequacy separate from convergence	Residual and ARCH diagnostics	Add posterior predictive scale checks
Forecast evaluation	Rolling forecasts from 15 to 120 minutes with fixed training-based estimates and sequential state updates	Propagate full posterior parameter uncertainty in a future extension

11 Limitations

1. The custom posterior target does not enforce nonseasonal AR stationarity and MA invertibility during sampling, although all retained draws passed the post hoc root audit.
2. The rolling evaluation uses fixed model-specific training estimates. For Bayesian models, posterior means are used rather than full posterior draws, so full posterior parameter uncertainty is not propagated.
3. The ARCH-LM test is applied to posterior-mean residuals and uses a specified 12-lag auxiliary regression.
4. Forecast evaluation uses one individual and one fixed holdout period. Rolling-origin replication is needed for stronger model comparison.

5. Timestamp-matched meals, insulin doses, exercise, and sleep are unavailable.
6. Thinning is used for convenience rather than validity. Unthinned sensitivity results should be retained in the reproducibility archive.
7. The paired CRPS analysis uses a fixed 12-origin moving-block length. Sensitivity to alternative block lengths should be retained in the reproducibility archive.
8. No multiplicity adjustment is applied across the 15 model-by-horizon comparisons, so the paired intervals are exploratory.
9. PIT diagnostics use non-overlapping targets to reduce dependence, but residual temporal dependence and shared parameter estimation may still affect the nominal Kolmogorov–Smirnov reference distribution.
10. All results in this report derive from a single result manifest (Table 12). Values appearing in earlier drafts of this project were produced under different MCMC and forecast versions and are superseded; they are retained in the reproducibility archive but are not comparable with the values reported here.
11. Prior sensitivity has not been assessed. The degrees-of-freedom prior $\log(\nu-2) \sim \mathcal{N}(\log 6, 1)$ should be re-run with at least one wider and one differently centred alternative, and the posterior mean and HDI for ν compared.
12. The four chains were initialized from modest perturbations around a common reference vector rather than from strongly overdispersed starting points. Consequently, the reported rank-normalized split $\hat{R}$ values should be interpreted with appropriate caution.

12 Reproducibility manifest

Table 12: Files and settings defining the official result version.

Item	Official value or path
Repository	https://github.com/sIVY0905/USRP
Training data	sarima_t_output/y_train.npy
Holdout data	sarima_t_output/y_test.npy
Training mean	sarima_t_output/center_value.npy
Student- t retained draws	sarima_t_output/transformed_fixed_after_warmup.npy
Unconstrained retained draws	sarima_t_output/kept_eta_fixed_after_warmup.npy
Gaussian retained draws	model_gaussian_sarima/natural_draws.npy
MCMC seed	20260720
Forecast simulation seed	20260812
Predictive draws per model-origin-horizon	1000
Last adaptation iteration	3950

The manifest should also contain software versions, file hashes, execution dates, and the names of every figure and metric file used in the final presentation.

13 Contributions and code provenance

The Bayesian SARIMA-*t* model, the adaptive block Metropolis–Hastings sampler, the posterior diagnostics of Sections 6–8, and the residual and ARCH analyses were implemented by the author.

The rolling multi-horizon evaluation of Section 9 uses `v8_horizon_figure.py`, an evaluation harness supplied during the internal methodological review of 12 August 2026. Figures 9 and 10 and Table 8 are outputs of that harness. The author re-executed the harness on 22 August 2026 and confirms that the reported values reproduce exactly; the execution log is archived as `rerun_log-20260822_public.txt`.

Analysis code, processed data arrays, retained posterior draws, derived metric files, and the public reproducibility log are archived in the project repository at <https://github.com/sIVY0905/USRP>.

14 Conclusion

The CGM implementation realizes the main computational principles of the companion MCMC theory report. The posterior target is explicit, inference uses multiple chains, adaptation is restricted to warmup, the retained phase uses a fixed proposal, and posterior summaries are supported by modern multi-chain diagnostics and Monte Carlo precision measures. Residual diagnostics then show why computational stability must be separated from statistical adequacy: heavy-tailed innovations improve robustness, but large residuals and conditional scale dynamics remain.

The rolling evaluation shows that forecast horizon is the dominant determinant of predictive performance. At 15 and 30 minutes, the fitted models produce nearly indistinguishable predictive scores. From 60 minutes onward, daily seasonal information becomes progressively more useful, while the nonseasonal ARMA model becomes less competitive. Fourteen of the fifteen paired moving-block-bootstrap intervals include zero; the only interval excluding zero is the disadvantage of the nonseasonal model at 120 minutes. Predictive performance deteriorates substantially with horizon, and the shared decline in coverage suggests a long-horizon calibration issue. Together with the ARCH-LM result, this pattern motivates future models with time-varying predictive scale and full propagation of posterior parameter uncertainty.

Declaration of generative AI and AI-assisted technologies

During the preparation of this report, generative AI and AI-assisted tools were used as supplementary tools for code development and debugging, manuscript organization, and language editing.

Microsoft Copilot was used primarily to assist with language revision, manuscript organization, and checking the clarity of mathematical and statistical explanations. Generative AI tools were also used during Python code development to assist with debugging and diagnostic checks.

The methodological choices, statistical analyses, interpretation of results, and conclusions were determined by the author. All statistical analyses were executed by the author in the local com-

puting environment, and the reported numerical results were verified against the corresponding program outputs. References and externally verifiable factual statements were checked against the cited sources. AI-generated suggestions were reviewed and revised by the author before being incorporated into the report.

The author takes full responsibility for the analysis, interpretation, reported results, code, and final content of this report.

A Remaining reproducibility checks

1. Verify and archive the official Student-*t* MCMC base seed and the chain-specific initial values or their exact generation rule.
2. Archive the exact version of `v8_horizon_figure.py` used to produce the final figure and metric files.
3. Record the forecast random seed used to generate the common uniform array.
4. Retain sensitivity results for alternative moving-block lengths.
5. Assess forecast-metric Monte Carlo variability across repeated simulation seeds.
6. Retain an unthinned MCMC sensitivity analysis.
7. Extend the rolling forecasts to integrate over full posterior parameter uncertainty.

B Superseded bootstrap specification

An earlier draft used an 8-origin block with 5000 replications. That specification is superseded. All paired CRPS intervals reported in the main analysis use a 12-origin moving block, corresponding to three hours, and 4000 bootstrap replications, as documented in Table 10.

C Superseded overlapping-target PIT calculation

The values in Table 13 were computed using all n_h overlapping forecast targets and are retained only to document the analysis version history. Because overlapping multi-step forecasts produce serially dependent PIT values, the usual independence assumption of the Kolmogorov–Smirnov reference distribution is not satisfied. These p -values are therefore not treated as valid calibration evidence and are superseded by the non-overlapping analysis reported in Section 9.4.

For Gaussian SARIMA at the 60-minute horizon, the overlapping calculation gives $p = 4.7 \times 10^{-5}$, whereas the non-overlapping calculation gives $p = 0.039$, a difference of approximately three orders of magnitude.

Table 13: Superseded overlapping-target PIT Kolmogorov–Smirnov p -values. Not valid calibration evidence; retained for version audit only.

Model	Horizon	Overlapping PIT KS p
Gaussian SARIMA	15 min	0.1024
Gaussian SARIMA	30 min	0.0034
Gaussian SARIMA	60 min	0.000047
Gaussian SARIMA	90 min	0.000044
Gaussian SARIMA	120 min	0.000067
Student- t SARIMA	15 min	0.1461
Student- t SARIMA	30 min	0.0055
Student- t SARIMA	60 min	0.000031
Student- t SARIMA	90 min	0.000019
Student- t SARIMA	120 min	0.000029
SARIMA MLE	15 min	0.0596
SARIMA MLE	30 min	0.0059
SARIMA MLE	60 min	0.000020
SARIMA MLE	90 min	0.000026
SARIMA MLE	120 min	0.000135
ARMA(2,2), no seasonal	15 min	0.0727
ARMA(2,2), no seasonal	30 min	0.0037
ARMA(2,2), no seasonal	60 min	0.000126
ARMA(2,2), no seasonal	90 min	0.000030
ARMA(2,2), no seasonal	120 min	0.000043

Bibliography

- [1] Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007. <https://doi.org/10.2307/1912773>.
- [2] Geweke, J. (1993). Bayesian treatment of the independent Student- t linear model. *Journal of Applied Econometrics*, 8(S1), S19–S40. <https://doi.org/10.1002/jae.3950080504>.
- [3] Haario, H., Saksman, E., and Tamminen, J. (2001). An adaptive Metropolis algorithm. *Bernoulli*, 7(2), 223–242. <https://doi.org/10.2307/3318737>.
- [4] Roberts, G. O., and Rosenthal, J. S. (2001). Optimal scaling for various Metropolis–Hastings algorithms. *Statistical Science*, 16(4), 351–367. <https://doi.org/10.1214/ss/1015346320>.
- [5] Vehtari, A., Gelman, A., Simpson, D., Carpenter, B., and Bürkner, P.-C. (2021). Rank-normalization, folding, and localization: An improved $\hat{R}$ for assessing convergence of MCMC. *Bayesian Analysis*, 16(2), 667–718. <https://doi.org/10.1214/20-BA1221>.
- [6] Zhu, J. (2022). Diabetes Datasets-ShanghaiT1DM and ShanghaiT2DM [Data set]. *Figshare*. <https://doi.org/10.6084/m9.figshare.20444397>.
- [7] Zhao, Q., Zhu, J., Shen, X., Lin, C., Zhang, Y., Liang, Y., Cao, B., Li, J., Liu, X., Rao, W., and Wang, C. (2023). Chinese diabetes datasets for data-driven machine learning. *Scientific Data*, 10, 35. <https://doi.org/10.1038/s41597-023-01940-7>.
- [8] Zhao, Q., Zhu, J., Wang, C., and Rao, W. (2022). Diabetes Datasets, ShanghaiT1DM and ShanghaiT2DM [Data set]. *figshare*. <https://doi.org/10.6084/m9.figshare.c.6310860>.